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Unit-2
Bipolar Junction Transistors (BJT)

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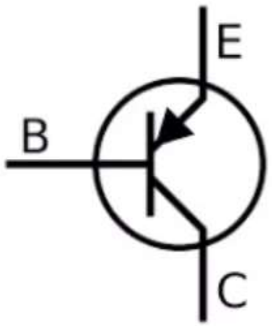
Bipolar Junction Transistor (BJT)

- ➢ Beside diodes, the most popular semiconductor devices is transistors. Eg: Bipolar Junction Transistor (BJT)
- ➢ Few most important applications of transistor are: as an amplifier and as a switch
- ➢ Transistor is a three terminal device namely Base, emitter and collector.
- ➢ In bipolar transistor the current conduction is due to both the types of charge carriers, namely holes and electrons. Hence this is called Bipolar Junction Transistor (BJT).
- ➢ In BJT the output current is controlled by input current and hence it is a current controlled device.

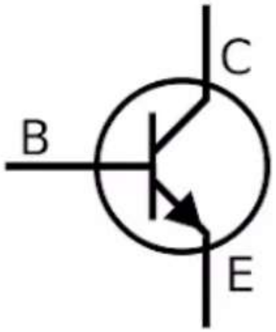
Types of BJT

- NPN Transistor
- PNP Transistor

Typical Bipolar Junction Transistor



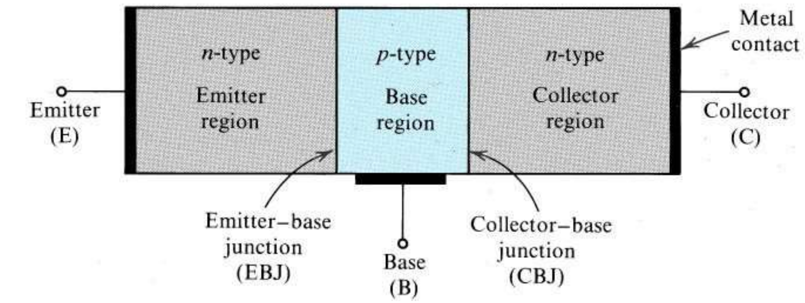
PNP BJT



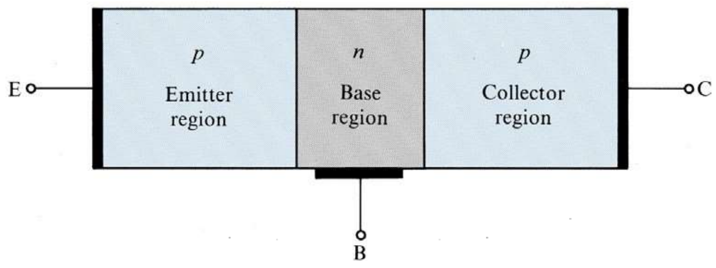
NPN BJT

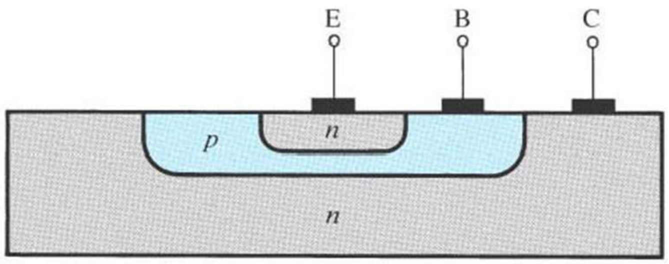
Bipolar Junction Transistor (BJT)

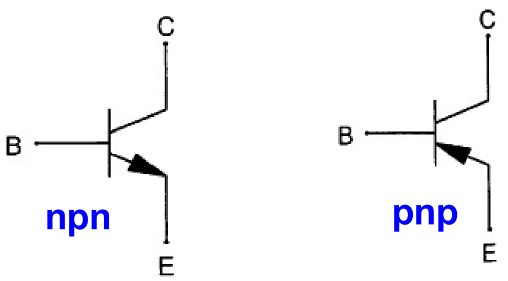
npnTransistor



pnpTransistor



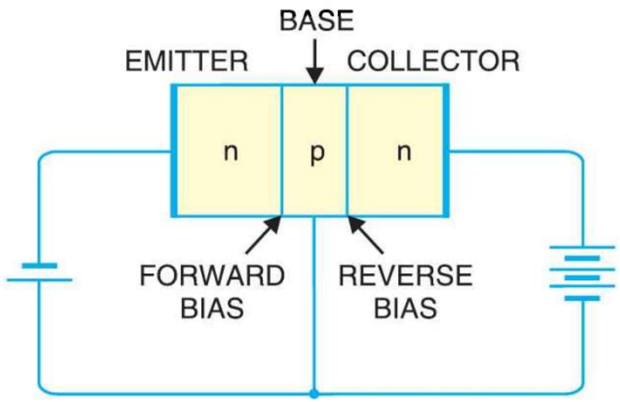




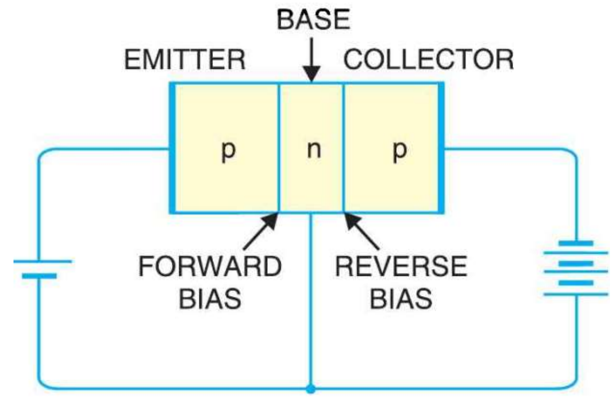
- ❖ BJT is a 3 terminal device. namely- emitter, base and collector
- ❖ npn transistor: emitter & collector are n-doped and base is p-doped.
- ❖ Emitter is heavily doped, collector is moderately doped and base is lightly doped and base is very thin.

Mode of operation for BJT

- In order to operate transistor properly as an amplifier, it is necessary to correctly bias the two p-n junctions with external voltages.
- Depending upon external bias voltage polarities used, the transistor works in one of the three regions, Active region, Cut-off region and Saturation region

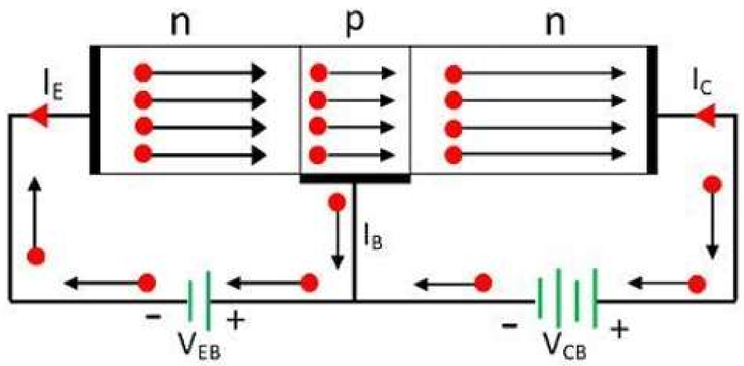


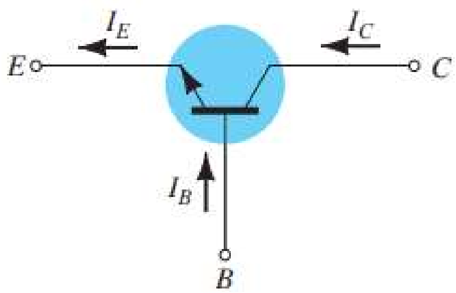
Region	Emitter-Base Junction (J_E)	Collector Base Junction (J_C)
Cut-off	reverse biased	reverse biased
Active	forward biased	reverse biased
Saturation	forward biased	forward biased



Working Principles of NPN Transistor

- The Emitter-Base (EB) junction is forward bias and Collector-Base (CB) is reversed bias.
- The forward biased EB junction causes the electrons in the n-type emitter to flow towards the base. This constitutes the emitter current I_E.
- *As* these electrons flow through the p-type base, they tend to combine with holes in p-region (base). Due to light doping, very few of the electrons injected into the base from the emitter recombine with holes to constitute base current, I_B.
- The remaining large number of electrons cross the base region and move through the collector region to the positive terminal of the external d.c source. This constitutes collector current I_C.
- Since, most of the electrons from emitter flow in the collector circuit and very few combine with holes in the base. Thus, the collector current is larger than the base current.

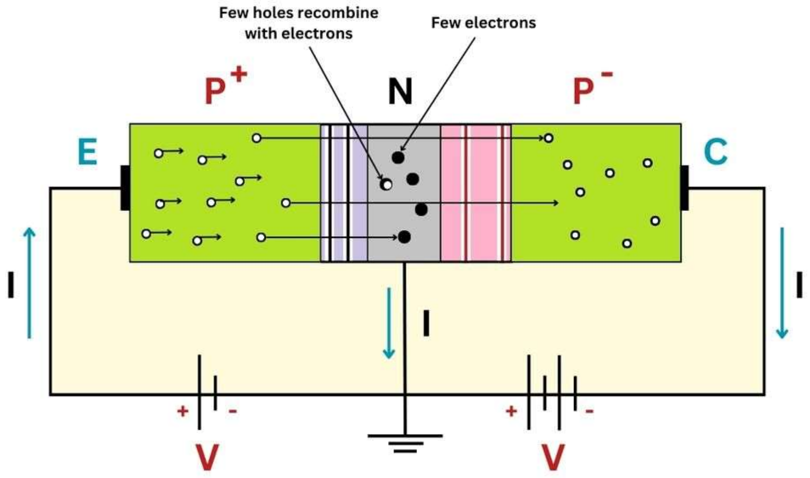




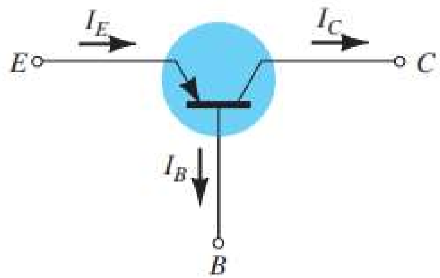
$$I_E = I_B + I_C$$

Working Principles of PNP Transistor

- The forward biased EB junction causes the holes in the p-type emitter to flow towards the base. This constitutes the emitter current IE.
- As these holes flow through the n-type base, they tend to combine with electrons in n-region(base). Since, the base is very thin and lightly doped, very few of the holes injected into the base from the emitter recombine with electrons to constitute base current, IB.

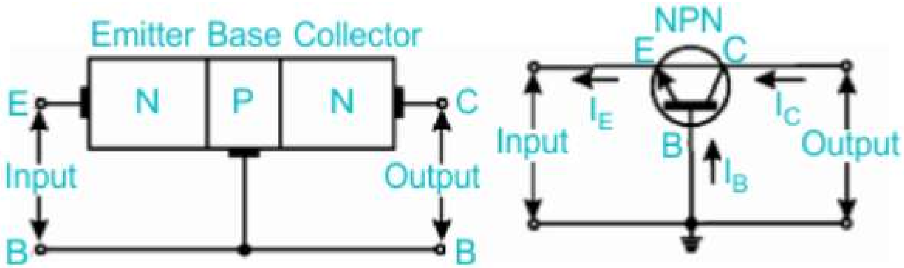


- The remaining large number of holes crosses the depletion region and move through the collector region to the negative terminal of the external d.c source. This constitutes collector current IC.
- Thus, the hole flow constitutes the dominant current in an p-n-p transistor. Since, most of the holes from emitter flow in the collector circuit and very few combine with electrons in the base. Thus, the collector current is larger than the base current.



$$I_E = I_B + I_C$$

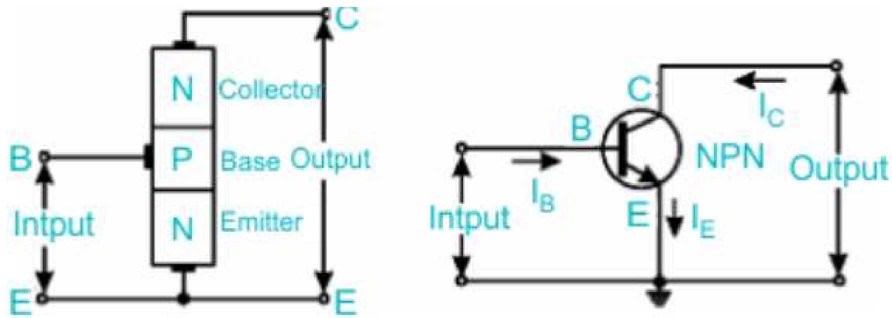
Different Configuration of BJT



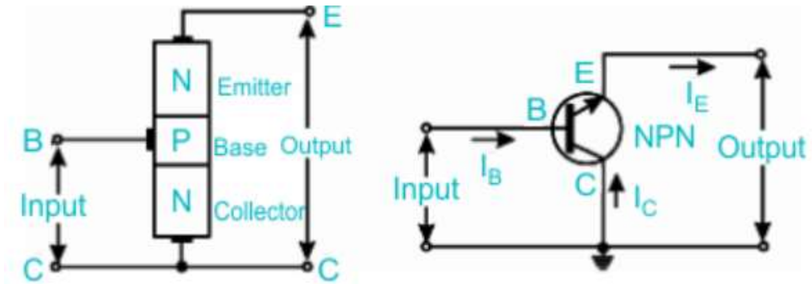
Common Base Configuration

•Common Base Connection (CB) Configuration
- Base is common to input and output
- High voltage gain, low input impedance, no current gain
- Used in high-frequency applications

•Common Emitter (CE) Configuration
- Emitter is common to input and output
- High voltage and current gain
- Most widely used for amplification



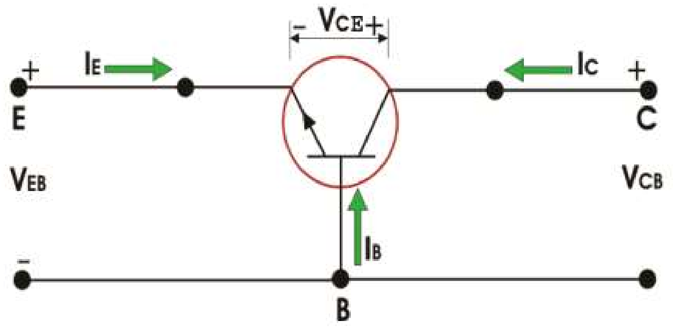
Common Emitter Configuration

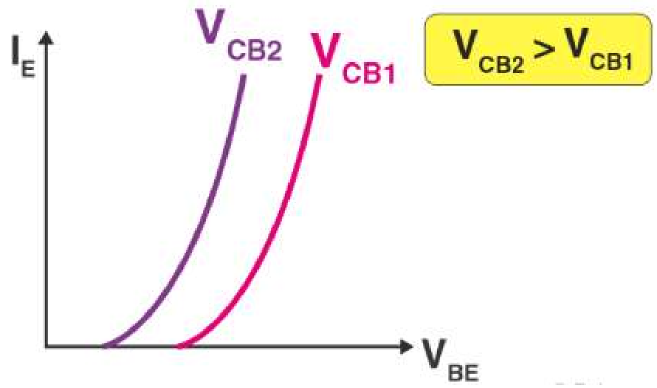


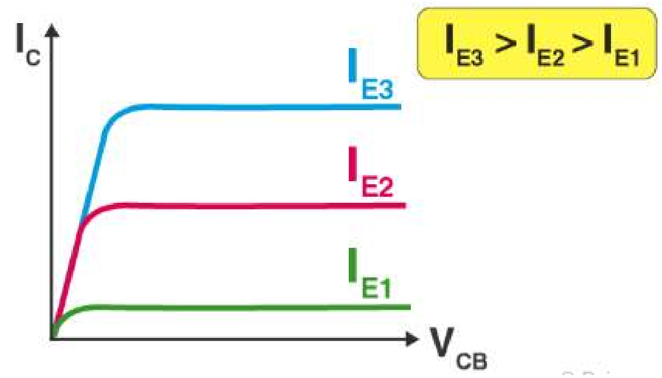
Common Collector Configuration

•Common Collector Configuration (CC) also called Emitter Follower
- Collector is common to input and output
- High input impedance, low output impedance, unity voltage gain
- Used for impedance matching

Input and Output Characteristics Common Base Configuration







In CB Configuration, the Base terminal of the transistor will be connected common between the output and the input terminals.

Input Characteristics: The variation of emitter current(I_E) with Base-Emitter voltage(V_{BE}), keeping Collector Base voltage(V_{CB}) constant.

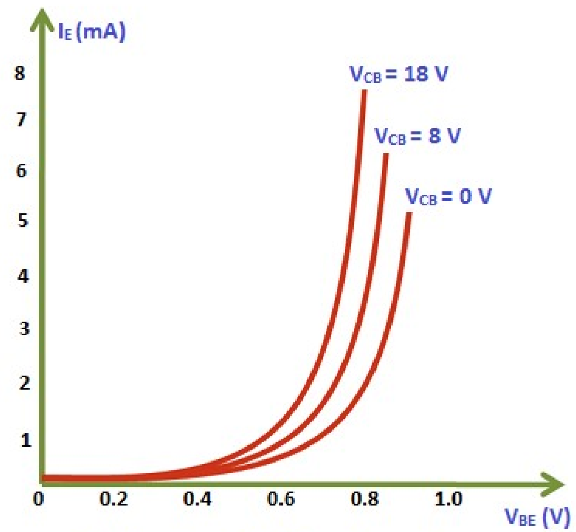
Output Characteristics: The variation of collector current(I_C) with Collector-Base voltage(V_{CB}), keeping the emitter current(I_E) constant.

Input Characteristics of CB Configuration

- The input characteristics of a BJT in common base configuration represent the relationship between the emitter current and the emitter–base voltage while keeping the collector–base voltage constant.
- Since the emitter–base junction is forward biased, an increase in emitter–base voltage reduces the barrier potential of the junction and allows more charge carriers to flow from the emitter into the base region.
- As a result, the emitter current increases rapidly with a small increase in emitter–base voltage.
- The input characteristic curve therefore resembles the forward bias characteristic of a PN junction diode.

Base Width Modulation (or) Early Effect

- When reverse bias voltage VCB increases, the width of depletion region also increases, which reduces the electrical base width.
- This reduces recombination in the base region, and hence a slightly smaller emitter–base voltage is required to produce the same emitter current.
- Due to this reason, the input characteristics shift slightly with changes in collector–base voltage.



I/p characteristics CB configuration

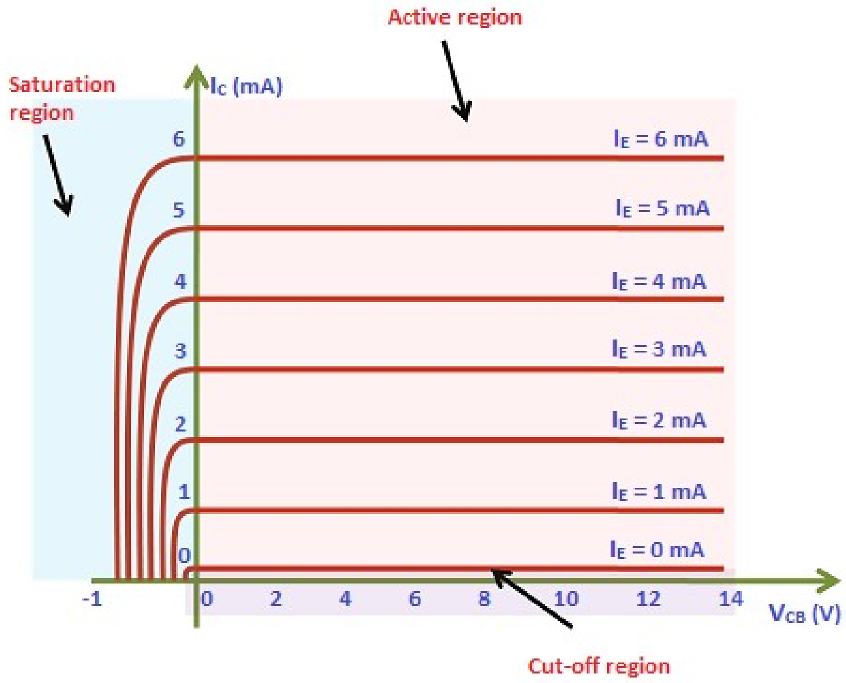
Output Characteristics of CB Configuration

The curve may be divided into three important regions namely saturation, active and cut off region.

Active Region

➢ The active region is the region between the vertical dashed line and the horizontal axis.

➢ In the active region, the collector current is constant and is equal to the emitter current.



O/P characteristics CB configuration

Saturation Region

- The saturation region is the region to the left of the vertical dashed line.
- It may be noted that in this region, collector to base voltage (VCB) is negative for a NPN transistor.
- In this region, a small change in VCB results in a large value of collector current.

Cut off Region

- The cut off region is the region along the horizontal axis as shown by a shaded region in the figure.
- It corresponds to the curve marked IE = 0.
- The collector current flows even when the collector to base voltage (VCB) is zero.
  A small collector current flows even when emitter current (IE) is zero.

Current Relation in Common Base Configuration

In common base configuration, the **current transfer mode ratio (α)** of a bipolar transistor in the forward active mode is defined as the ratio of the collector current (I_C) to the emitter current (I_E):

$$\boxed{\alpha = \frac{I_C}{I_E}}$$

I_C = Collector current

I_E = Emitter current

α < 1 (typically 0.95 – 0.99)

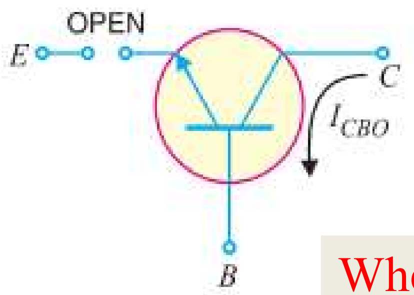
$$I_E = I_C + I_B$$

Leakage current: It is the **small unwanted current** that flows in a transistor **even when the input junction is not forward biased**, mainly due to **minority carriers** and it **increases with temperature**.

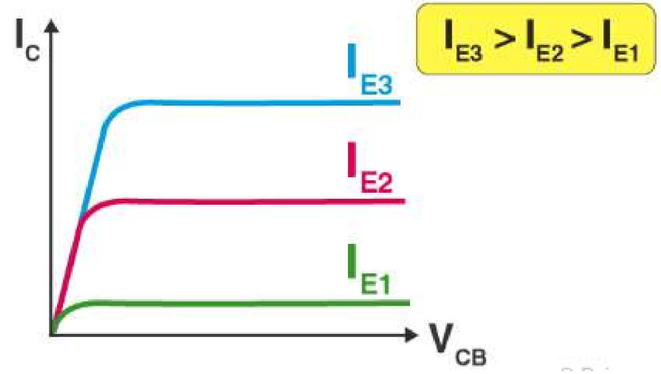
Leakage current: I_{CBO}

•I_{CBO} =Collector–Base leakage current

•Flows when **emitter is open** and **CB junction is reverse biased**

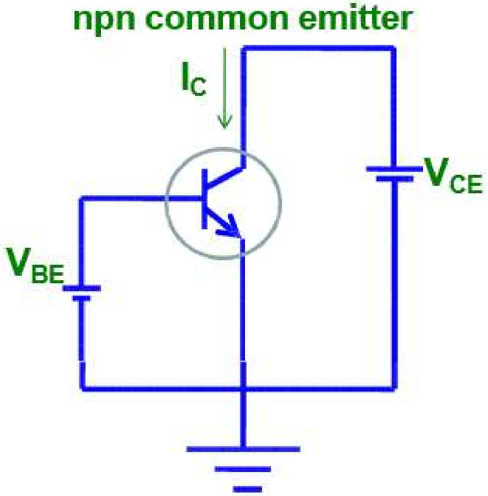


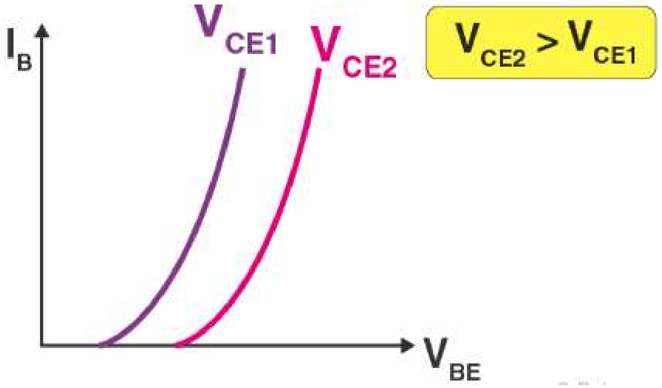
$$I_C = \alpha I_E + I_{CBO}$$



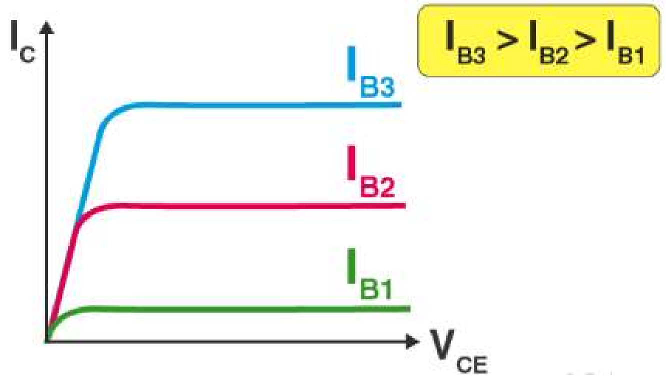
When I_E=0 , I_C=I_{CB0} is the current caused by the minority carriers crossing the pn-junction. I_{cB0} is leakage current called as collector base current with emitter open.

Input and Output Characteristics Common Emitter Configuration





Input Characteristics (CE)



Output Characteristics of CE

In CE Configuration, the Emitter terminal of the transistor will be connected common between the output and the input terminals.

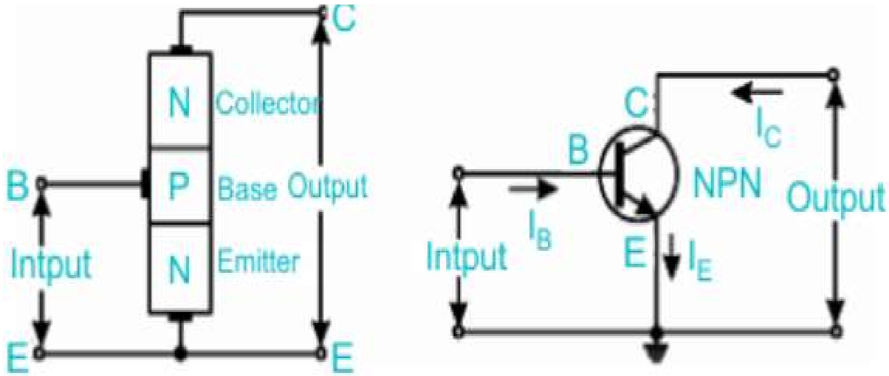
Input Characteristics (CE): The input characteristics of a CB transistor are the plot of base current (I_B) versus emitter–base voltage (V_{EB}) at a constant collector–emitter voltage (V_{CE})

Output Characteristics (CE): The output characteristics of a CB transistor are the plot of collector current (I_C)versus collector–emitter voltage (V_{CE}) a constant collector current (I_C)

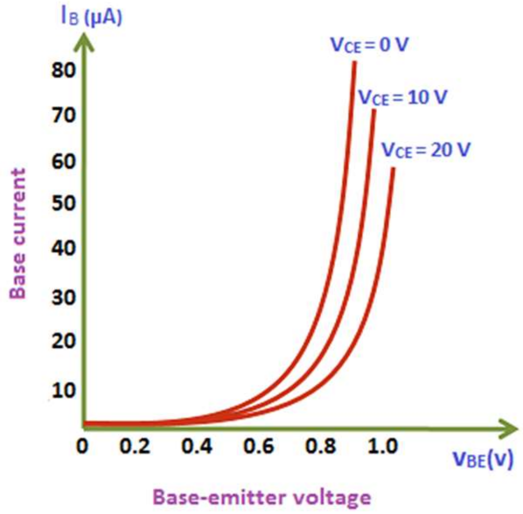
Input Characteristics of BJT in Comon Emitter Configuration

The input characteristics of a BJT in common emitter configuration describe the relationship between the base current and the base–emitter voltage while keeping the collector–emitter voltage constant.

- ➢ In the common emitter configuration, the base–emitter junction is forward biased, and therefore a small increase in the base–emitter voltage causes a significant increase in the base current.
- ➢ Due to this forward biasing, the input characteristic curve resembles that of a forward-biased PN junction diode.
- ➢ When the collector–emitter voltage is increased, the reverse bias across the collector–base junction increases, which reduces the effective base width due to the Early effect.
- ➢ As a result, for the same base current, a slightly smaller base–emitter voltage is required, causing a slight shift in the input characteristic curves.



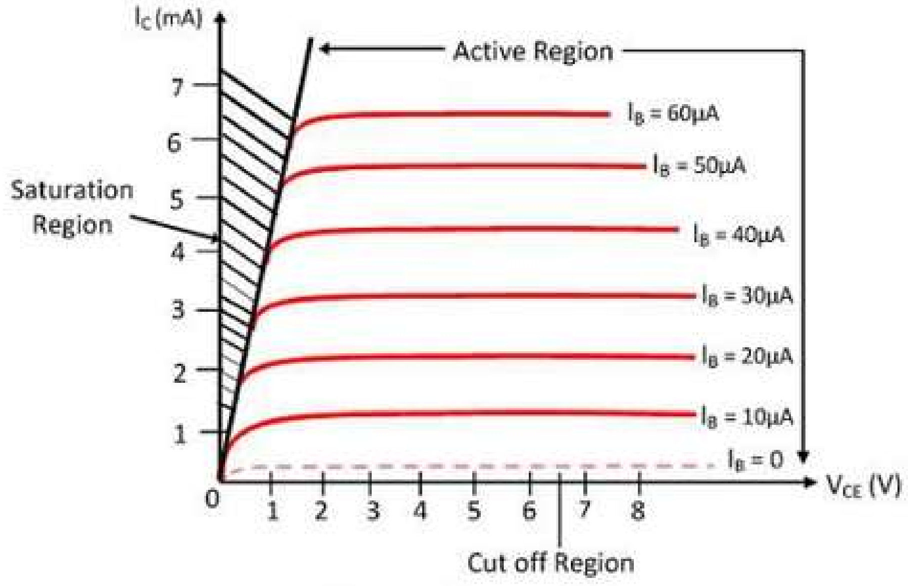
Common Emitter Configuration



I/P characteristics CE configuration

Output Characteristics of BJT in Comon Emitter Configuration

- The output characteristics of a BJT in common emitter configuration represent the relationship between the collector current and the collector–emitter voltage for a constant value of base current.
- In the active region of operation, the base–emitter junction is forward biased and the collector–base junction is reverse biased, and under these conditions, the collector current remains almost constant for a wide range of collector–emitter voltages.
- The collector current mainly depends on the base current and is approximately equal to the product of the base current and the current gain beta.
- At very low values of collector–emitter voltage, both junctions become forward biased and the transistor operates in the saturation region, where the collector current increases rapidly with voltage.



- When the base current is zero, the transistor operates in the cutoff region and only a small leakage current flows through the collector.